

Step by step process of normalization.

Consider the **Enrollments** relation below. It is not in any normal form.

Enrollments

Student ID	Student Name	Course ID	Course Title	Credits	Dept ID	Dept Name	Dept Office	Semester	Instructor	Grade	Textbooks (list)
S1	Aisha	C1	DBMS	5	D10	CS	B-201	Fall24	Smith	A	["DBMS Book", "Ramakrishnan"]
S1	Aisha	C2	AI	5	D10	CS	B-201	Fall24	Lee	B	["AI Book"]
S2	Bilal	C1	DBMS	5	D10	CS	B-201	Fall24	Smith	B+	["DBMS Book", "Ramakrishnan"]
S3	Chen	C3	Economics	4	D20	Econ	C-101	Fall24	Taylor	A-	["Microeconomics", "Macro Book"]

To be in 1NF, it should not have any multivalued attributes (Textbooks). Every cell should contain exactly one value.

Split the multivalued **Textbooks** into its own table.

So, we split it into two relations: **Enrollment_1NF** and **CourseTextbook**

Enrollment_1NF

Student ID	Student Name	Course ID	Course Title	Credits	Dept ID	Dept Name	Dept Office	Semester	Instructor	Grade
S1	Aisha	C1	DBMS	5	D10	CS	B-201	Fall24	Smith	A
S1	Aisha	C2	AI	5	D10	CS	B-201	Fall24	Lee	B
S2	Bilal	C1	DBMS	5	D10	CS	B-201	Fall24	Smith	B+
S3	Chen	C3	Economics	4	D20	Econ	C-101	Fall24	Taylor	A-

CourseID	TextbookTitle
C1	DBMS Book
C1	Ramakrishnan
C2	AI Book
C3	Microeconomics
C3	Macro Book

CourseTextbook

Key for Enrollment_1NF is (**StudentID, CourseID, Semester**) (one grade per course per semester).

2NF (remove partial dependencies on a *composite* key)

In **Enrollment_1NF**, non-key attributes depending only on part of the key:

- **StudentID** → **StudentName**
- **CourseID** → **CourseTitle, Credits, DeptID**
(Composite key is (StudentID, CourseID, Semester).)

Decompose:

- **Student(StudentID, StudentName)**
- **Course(CourseID, CourseTitle, Credits, DeptID, DeptName, DeptOffice)**
- **Enrollment(StudentID, CourseID, Semester, Instructor, Grade)**
- **CourseTextbook(CourseID, TextbookTitle)** (unchanged)

Why: now every non-key attribute in **Enrollment** depends on the **whole** key (no attribute depends only on StudentID or CourseID).

3NF (remove transitive dependencies)

Inside **Course**, we still have:

- **DeptID** → **DeptName, DeptOffice** (transitive: CourseID → DeptID → DeptName/DeptOffice)

Decompose:

- **Department**(DeptID, DeptName, DeptOffice)
- **Course**(CourseID, CourseTitle, Credits, DeptID)

Now all relations are in **3NF**:

- **Student** - (StudentID (PK), StudentName)
- **Department** – (DeptID (PK), DeptName, DeptOffice)
- **Course** – (CourseID (PK), CourseTitle, Credits, DeptID)
- **Enrollment** – (StudentID, CourseID, Semester, Instructor, Grade) - PK (composite): (StudentID, CourseID, Semester)
- **CourseTextbook** – (CourseID (FK → Course.CourseID), TextbookTitle)

Usually, relations in 3NF are good enough. Unless our particular business rules create some anomalies. In that case, BCNF is required.

Example Rule: Each instructor teaches exactly **one** CourseID.

Functional dependency: **Instructor** → **CourseID**.

Check **Enrollment(StudentID, CourseID, Semester, Instructor, Grade)**:

- Known FD: **(StudentID, CourseID, Semester) → Instructor, Grade** (by design).
- Business FD: **Instructor → CourseID** (not a superkey).

Why 3NF allows it

RHS (**CourseID**) is a **prime** attribute (part of the candidate key (StudentID, CourseID, Semester)), so 3NF is satisfied.

Why BCNF rejects it

BCNF demands LHS be a **superkey**. **Instructor** is **not** a superkey of Enrollment → violates BCNF.

Decompose Enrollment to reach BCNF

Split the overlapping facts:

1. **InstructorCourse(Instructor, CourseID)**
(captures the rule: an instructor determines their course)
2. **StudentTeaching(StudentID, Instructor, Semester, Grade)**
(who teaches each student in a given semester, with the grade)

Resulting BCNF set

- **Student**(StudentID, StudentName)
- **Department**(DeptID, DeptName, DeptOffice)
- **Course**(CourseID, CourseTitle, Credits, DeptID)
- **CourseTextbook**(CourseID, TextbookTitle)
- **InstructorCourse**(Instructor, CourseID) $\leftarrow$ BCNF
- **StudentTeaching**(StudentID, Instructor, Semester, Grade) $\leftarrow$ BCNF

Exercise: Consider the relation below. Show step by step process to normalize this table up to BCNF.

EmpID	EmpName	DeptID	DeptName	DeptOffice	ProjectID	PName	Manager	Duration	Skills
E1	Ali	D10	IT	B201	P1	Payroll	Smith	6 m	[Java, SQL]
E1	Ali	D10	IT	B201	P2	Website	Lee	4 m	[HTML, CSS, JS]
E2	John	D10	IT	B201	P1	Payroll	Smith	6 m	[Java, SQL]
E3	Sara	D20	HR	C101	P3	Recruitment	Taylor	3 m	[Excel]